Huy H. Nguyen Nguyễn Hồng Huy

AI Safety · Multimedia Security · Trustworthy Generative AI

Researcher, SB Intuitions (a SoftBank Group company)
Visiting Associate Professor, National Institute of Informatics (NII)

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Researcher in AI safety and multimedia security, developing **guardrails and safety evaluations for large language, vision-language, and text-to-image models**, building on earlier work in deepfake detection, media forensics, and adversarial machine learning. Author of **70+ papers** at venues including WACV, EMNLP, AACL-IJCNLP, ICCV, ACM MM, ICASSP, and IJCB, with **3,600+ citations**; guest editor, workshop organizer, and tutorial speaker on AI safety.

3,600+	20	72	2×
citations (Google Scholar)	h-index	publications	BTAS/IJCB 5-Year Highest Impact Award

RESEARCH INTERESTS

AI safety and guardrails for LLMs and VLMs · **safety evaluation of generative models** · deepfakes and multimedia forensics · adversarial machine learning and robustness · misinformation and AI-generated content · privacy and biometrics

POSITIONS

2025.04 – Present — **SB Intuitions (a SoftBank Group company), Japan**
Researcher, R&D Division – Responsible AI Team
Focus: guardrails and safety evaluation for LLMs, VLMs, and text-to-image models (WACV 2027, EMNLP 2026, AACL-IJCNLP 2026); AI safety tutorial at NLP 2026.

2022.04 – Present — **National Institute of Informatics (NII), Japan**
Visiting Associate Professor (2025.04 – Present)
Project Assistant Professor (2022.11 – 2025.03); Project Postdoctoral Researcher (2022.04 – 2022.10)
Focus: deepfakes, adversarial machine learning, and misinformation; Principal Investigator, JST CREST AIP Challenge (2022–2023); technical project NII SYNTHETIQ VISION.

2020.07 – 2020.11 — **Amazon.com, Inc. (virtual summer internship)**
Amazon Selection and Catalog Systems team; built a toolset to assist AutoML session debugging.

2019.10 – 2020.02 — **Idiap Research Institute, Switzerland (research internship)**
Biometrics Security and Privacy group; advisor: Dr. Sébastien Marcel. Master faces (IJCB 2020, T-BIOM 2022).

2015.02 – 2015.08 — **National Institute of Informatics, Japan (research internship)**
Echizen Laboratory; advisor: Prof. Isao Echizen. Image spoofing attack detection (IWDW 2015).

2013.07 – 2015.02 — **Universal Technology Services Corporation, Vietnam**
Research and development: OPSWAT Metascan Splunk app, VoxyPAD.

2012.07 – 2012.10 — **Pohang University of Science and Technology (POSTECH), South Korea (internship)**
High Performance Computing Lab; advisor: Prof. Jong Kim. Secure and fault-tolerant Java coding guidelines.

EDUCATION

2016 – 2022 — **Ph.D., The Graduate University for Advanced Studies, SOKENDAI, Japan**
School of Multidisciplinary Sciences, Department of Informatics. Echizen Lab and the Global Research Center for Synthetic Media, NII. Advisors: Prof. Junichi Yamagishi and Prof. Isao Echizen.

2009 – 2013 — **B.Sc. (Honors Program), VNUHCM – University of Science, Vietnam**
Faculty of Information Technology. Thesis: applying biological sequence alignment to detect and classify computer viruses; advisor: Prof. Thuc D. Nguyen.

SELECTED AWARDS

2023, 2024 — **BTAS/IJCB 5-Year Highest Impact Award (IJCB 2023 and IJCB 2024)**
"Multi-task Learning for Detecting and Segmenting Manipulated Facial Images and Videos," BTAS 2019.

2025 — **Best Perspective Paper Award, APSIPA ASC 2025**
"Foundation Models as Guardrails: LLM- and VLM-Based Approaches to Safety and Alignment."

2023 — **Telecom Interdisciplinary Research Award**
Telecommunication Advancement Foundation, Japan.

2022 — **Best Student Award**
National Institute of Informatics, Japan.

2022 — **Excellent Paper Award, IEICE**
Three journal papers, Institute of Electronics, Information and Communication Engineers, Japan.

2018 — **Ph.D. Forum Best Poster Award, APSIPA ASC 2018**
"Real or Fake Images: Attacking and Reinforcing the Machine Learning Systems."

2017 — **Best Paper Award, IEEE WIFS 2017**
"An Approach for Gait Anonymization using Deep Learning."

RESEARCH GRANTS

2022 – 2023 — **JST CREST AIP Challenge: Deepfake Detection for eKYC Systems**
Principal Investigator

2020 – 2025 — **JST CREST: FakeMedia project**
Student, later staff member

2018 – 2023 — **JST-ANR: VoicePersonae project**
Student, later staff member

2016 – 2021 — **JSPS Kakenhi (S): Communication System for Defending against Attacks of Media Clones**
Student member

SELECTED PUBLICATIONS

Full list of 72 papers: honghuy127.github.io/publications · Citation counts: Google Scholar, October 2026

AI safety and guardrails

WACV 2027 — M.-D. Vo, J. Li, **H. H. Nguyen**, P. Kavumba, and K. Wataoka, "SafeGrad: A Severity-Graded Benchmark for Safety Evaluation, Defense, and Attack of Text-to-Image Models," *Winter Conference on Applications of Computer Vision (WACV) 2027*.

AACL-IJCNLP 2026 — **H. H. Nguyen**, P. Kavumba, J. Li, M.-D. Vo, and K. Wataoka, "The Guardrail Trinity: Procedural Safety Schemas, Reasoning Chain Decomposition, and Dual-Mode Inference for LLM Guardrails," *Asia-Pacific Chapter of the Association for Computational Linguistics and International Joint Conference on Natural Language Processing (AACL-IJCNLP) 2026, Main*.

EMNLP 2026 — J. Li, J. Zhang, M.-D. Vo, **H. H. Nguyen**, P. Kavumba, and K. Wataoka, "Do VLMs Share Safety Neurons Across Modalities?" *Empirical Methods in Natural Language Processing (EMNLP) 2026, Main Track*.

APSIPA ASC 2025 — **H. H. Nguyen**, P. Kavumba, T. Kurosawa, and K. Wataoka, "Foundation Models as Guardrails: LLM- and VLM-Based Approaches to Safety and Alignment," *Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC) 2025*.
Best Perspective Paper Award

Deepfakes and media forensics

APSIPA Trans. 2025 — **H. H. Nguyen**, S. Liang, J. Yamagishi, and I. Echizen, "Navigating Real and Fake in the Era of Advanced Generative AI," *APSIPA Transactions on Signal and Information Processing, vol. 14, no. 3 (2025)*.

IJCB 2024 — **H. H. Nguyen**, J. Yamagishi, and I. Echizen, "Exploring Self-Supervised Vision Transformers for Deepfake Detection: A Comparative Analysis," *International Joint Conference on Biometrics (IJCB) 2024*.

IJCB 2023 — **H. H. Nguyen**, J. Yamagishi, and I. Echizen, "How Close are Other Computer Vision Tasks to Deepfake Detection?" *International Joint Conference on Biometrics (IJCB) 2023*.

ICCV 2021
180+ citations
T.-N. Le, **H. H. Nguyen**, J. Yamagishi, and I. Echizen, "OpenForensics: Large-Scale Challenging Dataset for Multi-Face Forgery Detection and Segmentation In-The-Wild," *International Conference on Computer Vision (ICCV) 2021*.

BTAS 2019
740+ citations
H. H. Nguyen, F. Fang, J. Yamagishi, and I. Echizen, "Multi-task Learning for Detecting and Segmenting Manipulated Facial Images and Videos," *International Conference on Biometrics: Theory, Applications and Systems (BTAS) 2019*.
BTAS/IJCB 5-Year Highest Impact Award, 2023 & 2024

ICASSP 2019
1,200+ citations
H. H. Nguyen, J. Yamagishi, and I. Echizen, "Capsule-Forensics: Using Capsule Networks to Detect Forged Images and Videos," *International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2019*.

Adversarial machine learning, biometric security, and misinformation

WACV 2023
H. H. Nguyen, T.-N. Le, J. Yamagishi, and I. Echizen, "Analysis of Master Vein Attacks on Finger Vein Recognition Systems," *Winter Conference on Applications of Computer Vision (WACV) 2023*.

WACV 2023
F. Waseda, S. Nishikawa, T.-N. Le, **H. H. Nguyen**, and I. Echizen, "Closer Look at the Transferability of Adversarial Examples: How They Fool Different Models Differently," *Winter Conference on Applications of Computer Vision (WACV) 2023*.

T-BIOM 2022
H. H. Nguyen, S. Marcel, J. Yamagishi, and I. Echizen, "Master Face Attacks on Face Recognition Systems," *IEEE Transactions on Biometrics, Behavior, and Identity Science (2022)*.

IJCB 2020
H. H. Nguyen, J. Yamagishi, I. Echizen, and S. Marcel, "Generating Master Faces for Use in Performing Wolf Attacks on Face Recognition Systems," *International Joint Conference on Biometrics (IJCB) 2020*.

AINA 2020
220+ citations
D. I. Adelani, H. Mai, F. Fang, **H. H. Nguyen**, J. Yamagishi, and I. Echizen, "Generating Sentiment-Preserving Fake Online Reviews Using Neural Language Models and Their Human- and Machine-based Detection," *International Conference on Advanced Information Networking and Applications (AINA) 2020*.

INVITED TALKS, TUTORIALS, AND PANELS

2026.07
Panelist: "Building AI Readiness: Safety, Talent, and International Cooperation"
APO GAIA Public Symposium, University of Tokyo.

2026.03
Tutorial: "AI Safety: Bridging Academic Research and Industrial Application"
NLP 2026, Association for Natural Language Processing; with K. Wataoka and P. Kavumba.

2025.11
Perspective & panel session: "Foundation Models as Guardrails"
APSIPA ASC 2025, with K. Wataoka.

2024
"Balancing Innovation and Security: Addressing Deepfakes and Disinformation in the Age of Advanced AI"
N-CRiPT Workshop, National University of Singapore.

2021
"Deepfake Generation and Detection – An Overview"
Symposium on Sensing via Image Information (SSII), Japan.

PROFESSIONAL SERVICE

Guest editor
APSIPA Transactions on Signal and Information Processing, Time Series on "Deepfakes, Unrestricted Adversaries, and Synthetic Realities in the Generative AI Era."

Organizer
Workshop on Large Vision–Language Model Learning and Applications (LAVA)
ACCV 2024 and ACM MM 2025.

Session chair
ACCV 2024 Workshop on Large Vision–Language Model Learning and Applications (LAVA).
APSIPA ASC 2023 Special Session: Multimedia Security and Privacy in the Age of Deep Learning.
APSIPA ASC 2020 Special Session: Deep Generative Models for Media Clones and Its Detection.

Reviewer
Conferences: NeurIPS, ICLR, ICML, CVPR, ECCV, WACV, ICME, ACL Rolling Review, APSIPA ASC.
Journals: IEEE (T-PAMI, TIP, T-IFS, Access), IEEE/CAA JAS, ACM TOMM, Elsevier (Pattern Recognition Letters, EAAI), EURASIP JIVP, IEICE Transactions.

TEACHING

2026
Instructor, APSIPA ASC 2026 Winter School 2: Foundation Models and Agentic AI

2013.09 – 2016.06
Teaching Assistant, VNUHCM – University of Science, Vietnam
Data Structures and Algorithms; Network Security; Computer Architecture and Assembly; Computer Vision.